

Design and Implementation of a Multi-Interface Serial Communication System

Abstract— This paper presents the design and implementation of a multi-interface serial communication system integrating PCIe, USB, and Ethernet interfaces. The system enables connection-agnostic communication, ensuring reliable and scalable data transfer across different physical layers. The design focuses on latency optimization, driver compatibility, and system scalability for industrial and embedded applications.

1. Introduction

This report explores modern serial communication systems where devices are no longer limited by physical interfaces. The system enables communication over PCIe, USB, and Ethernet using a unified architecture.

2. System Architecture

The system consists of a UART controller interfaced with PCIe, USB, and Ethernet modules. A host system accesses the communication through standard APIs and virtual COM ports.

3. Hardware Design

The hardware includes a PCIe expansion card with USB 3.0 and Gigabit Ethernet support, UART controllers with FIFO buffers, and supporting components such as oscillators and power connectors.

4. Software Design

The software layer includes device drivers, OS APIs, and application-level communication. Drivers ensure compatibility and efficient communication between hardware and applications.

5. Performance Analysis

Latency varies across interfaces: PCIe offers the lowest latency, USB introduces moderate delay, and Ethernet provides scalable remote communication with acceptable latency.

6. Implementation

The implementation involves installing hardware, configuring drivers, assigning COM ports, and developing applications using standard APIs or libraries like PySerial.

7. Challenges

Challenges include latency variation, driver compatibility, and network delays. These are mitigated using adaptive timeouts and optimized drivers.

8. Conclusion

The proposed system provides a flexible and scalable communication solution. Ethernet is recommended for long-term deployment, while PCIe is ideal for low-latency applications.

Performance Comparison

Interface	Latency	Use Case
PCIe	Low (~1 ms)	Real-time control
USB	Medium (16–75 ms)	Peripheral connectivity
Ethernet	Variable (5–20 ms)	Remote communication